

Tutorial of binding affinity prediction by MolAICal and OnionNet model

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1. Introduction

Deep learning model can be used to evaluate binding affinity between ligands and receptors. In this tutorial, OnionNet [1] is introduced to calculate binding affinity of ligands by MolAICal. This example can be for users to know quickly how deep learning predicts binding affinity between ligands and proteins. This tutorial can only run on Linux system. For more detailed MolAICal, please visit: <https://molaical.github.io>.

2. Materials

2.1. Software requirement

1) MolAICal: <https://molaical.github.io>

2.2. Example files

1) All the necessary tutorial files are downloaded from:

<https://github.com/MolAICal/tutorials/tree/master/014-bindingaffinityOnionnet>

3. Procedure

3.1 Install OnionNet model

Currently, only Linux version of MolAICal is supported.

1) Click and open website: [DownloadModel](#)

Then go to the folder “science--AImodels--BindingAffinity--onionnet--linux” in the open website, download a file named “onionnet.tar.gz”

2) Move “onionnet.tar.gz” to the folder “MolAICal-xxx/mtools” in the MolAICal container. Where “MolAICal-xxx” is your decompressed root directory of MolAICal in the MolAICal container. The folder “mtools” is the targeted directory. The movement procedures are as follows:

a) First of all, copy the files to the MolAICal container.

```
***** 1. Go to the container file system, it will enter the "/root" *****  
#> molaical.exe -eset shell in  
  
***** 2. The first part of 'cp' is from the local machine, and the second part is in the container *****  
#> mv <Your machine path>/onionnet.tar.gz /root/soft/MolAICal-xxx/mtools  
  
***** 3. Exit container file system *****  
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical  
#> cd /root/soft/MolAICal-xxx/mtools
```

Note: 'molaical' is the container name.

3) Decompress file

```
#> tar -xvzf onionnet.tar.gz
```

And go to folder named OnionNet:

```
#> cd onionnet
```

4) install OnionNet model

```
#> chmod +x install.sh
```

```
#> ./install.sh
```

Until now, OnionNet model is installed completely.

✧ Remember to use the "exit" command to quit the MolAICal virtual environment and return to the local computer for computation (primarily to avoid file-copying steps; execution within the container is also possible but requires copying data from the local computer to the MolAICal container).

```
#> exit
```

3.2 Calculate binding affinity for one complex

3.2.1 Merge one protein and one ligand into a complex file

[Open 014-bindingaffinityOnionnet](#)

Then use the below command to prepare complex:

```
#> molaical.exe -model mergeon -r 1a30_protein.pdb -l 1a30_ligand.mol2.pdb -c 1a30_complex.pdb
```

Notice: if users do not have pdb format molecular file, they use the command of MolAICal to convert molecules in pdb format. For instance, users have mol2 format file, they can use the below command for format conversion (**Note:** molecule should have correct suffix so that MolAICal can recognize them automatically):

```
#> molaical.exe -tool format -i ligand.mol2 -o ligand.pdb
```

3.2.2 Calculate pK_x (pK_a or pK_i) for binding affinity prediction

Put the merged complex files into a file. In this tutorial, this file named "inputlist.dat" contains complex file name. Then, using the below command for pK_x calculation:

```
#> molaical.exe -model onionnet -i inputlist.dat -o results.csv
```

It will generate a file named results.csv that contains pK_x value of one complex. If users want to convert pK_x to binding free energy, they can refer to "<https://molaical.github.io/tutorial.html>" and MolAICal manual which part is involved in "Binding free energy from pK_d or pK_i".

3.3 Calculate binding affinity for many complexes

3.3.1 Merge many ligands one protein into many complex files

Sometimes, users want to calculate binding affinities of many ligands, under these circumstances, it needs to merge many ligands and one protein into many complex files.

[Open 014-bindingaffinityOnionnet/list](#)

Putting all ligand names into a file called “list.txt”, in this part, the protein file is named to GCGRNoLigand.pdb. Then use the below command to merge one protein and many ligand files:

```
#> molaical.exe -model mergeon -r GCGRNoLigand.pdb -f list.txt
```

It will produce files named “complex_<you ligand name>.pdb”.

3.3.2 Calculate pKx (pKa or pKi) for binding affinity prediction

Put all merged complex files into a file called “complexList.txt” using below command:

```
#> ls -A complex_*.pdb > complexList.txt
```

and then using the below command to calculate binding affinity:

```
#> molaical.exe -model onionnet -i complexList.txt -o results.csv
```

It will generate a file named results.csv that contains pKx values of many complexes. If users want to convert pKx to binding free energy, they can refer to “<https://molaical.github.io/tutorial.html>” and MolAICal manual which part is involved in “Binding free energy from pKd or pKi”.

Reference

1. Zheng L, Fan J, Mu Y. OnionNet: a Multiple-Layer Intermolecular-Contact-Based Convolutional Neural Network for Protein-Ligand Binding Affinity Prediction. ACS Omega. 2019;4(14):15956-65. Epub 2019/10/09. doi: 10.1021/acsomega.9b01997. PubMed PMID: 31592466; PubMed Central PMCID: PMC6776976.